

Optimization of centroid-finding algorithms for cathode strip chambers[☆]

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Abstract

We compare the data obtained by a precision cathode strip chamber (CSC) with a static point charge model. The observed strip charge profile is in good agreement with the model prediction. We have used this model to calculate the systematic error of some popular centroid-finding algorithms, and found good agreement with the systematic error observed in the data. We further studied the effect of incidence angle and of anode wire displacements. By using an algorithm based on a hyperbolic secant function (sech^2) to describe the charge profile, we are able to improve the published single layer resolution of our CSC prototype in a muon beam by 14%, from 53.3 ± 0.9 to $45.8 \pm 0.7 \mu\text{m}$.

1. Introduction

Recently, there is a resurgence of interest in cathode strip chambers (CSCs), a wire chamber type in which the position of an avalanche is measured along the anode wire by sampling the charge profile induced on the cathode [1–3]. The cathode of a CSC is segmented as strips oriented perpendicular to the anode wire. The width of the strips is usually of the order of the wire-cathode distance. The avalanche could be generated by a MWPC or a drift tube. Fig. 1 shows the schematic for an open profile drift tube CSC, a design

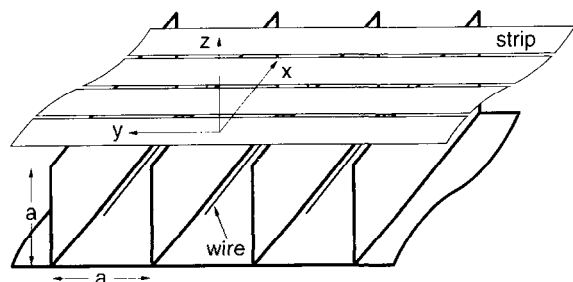


Fig. 1. Schematic of an open profile drift tube CSC and the coordinate system in this paper.

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studied in this paper, and the coordinate system used.

The renewed interest in CSCs is due to the need to instrument large areas in hadron collider detectors at the SSC or the LHC for muon detection [4–6]. The momentum of the muons expected in the hadron collider is high, resulting in a need for high spatial resolution, typically under $100 \mu\text{m}$ per layer. The CSC approach is considered to be more cost-effective than technologies based on drift time measurement because the precision of a CSC depends less critically on the placement accuracy of the anode wires. The stringing of thousands of anode wires to a typical precision of $20 \mu\text{m}$ is both time-consuming and expensive. The precision of a CSC is tied to the accuracy of the cathode strips. It has been demonstrated recently that precision strips of large area can be manufactured inexpensively [1,3]. Tests of several prototypes have shown that the achievable resolution is below $50 \mu\text{m}$ [1,3], which more than meets the requirement for most hadron collider muon detection. This is a dramatic improvement over previous attempts based on open profile drift tubes [7]. One of the main reasons for this success is the considerable progress made in the understanding and control of systematic effects limiting the resolution of CSCs.

Based on the charges measured in each strip, one calculates a charge centroid by using an algorithm, which should reproduce as closely as possible the position of the avalanche along the wire. It is a well known fact that most algorithms do not predict centroids that coincide with the actual avalanche position:

there is a systematic difference between the algorithmic centroid and the true avalanche position [8–10]. However, the systematic deviation can be modeled and corrected for.

In this paper we develop a simple model for the induced charge, and compare the simulated charge profile with data taken with a three layer CSC prototype [3] which has achieved a single layer resolution under $50\text{ }\mu\text{m}$. We examined some of the commonly used algorithms and found very good agreement between the systematic error predicted by our model and the systematic error seen in the data. We were able to find an optimized algorithm which improves the resolution of our prototype from $53.3 \pm 0.9\text{ }\mu\text{m}$ to $45.8 \pm 0.7\text{ }\mu\text{m}$. We also studied the effects of non-normal track incidence and of wire to strip distance variations.

The paper is organized as follows. A model for the charge induction is presented in Section 2. We describe the prototype and the data in Section 3, and compare the general features of the model with the data. In Section 4 we introduce the various algorithms and evaluate their performance using data and model. The general technique of systematic correction using a Fourier series is described in Section 5. The results are summarized in Section 6. The sensitivity of the algorithms to effects such as incidence angle and wire displacement is compared in Section 7.

2. The model

2.1. Introduction

In a typical wire chamber, the ionization electrons left behind a charged particle are drifted to the anode wire. The total number of electrons increases by a factor of three to four during the drift, but the initial position of the ionization is preserved. Charge multiplication of the order of 10 000 to 100 000 takes place near the anode wire where the electric field is high. After the rapid clearing of the electrons (in less than 10 ns) the positive ions are exposed, and an image charge is induced on the cathode. The ions are drifted slowly toward the cathode. Since the time-scale of the ion motion is long compared to the typical signal collection time for most CSC applications, the time-dependence of the image charge can be ignored. It is therefore a good approximation to consider the collective effect of the ions to be that of a point charge located at the anode wire. The total charge, of course, is subject to various stochastic fluctuations; the dominant one is the Landau fluctuation in the ionization process. One of the virtues of the CSC technique is that the centroid of the induced charge profile is independent of the absolute charge. Therefore, to

model the image charge of a CSC, it suffices to consider the case of a fixed point charge q at the wire.

2.2. Charge profile

The induced charge profile for any chamber geometry can be solved by standard techniques in electrostatics [11]. The solution is particularly simple when the chamber is a drift tube of square cross section, as in the case of the CSC studied in this paper, which can be solved by the image charge method. The four sides of the chamber are considered to be highly conducting and are held at ground potential. The problem can be reduced to two dimensions when the cell size a is small compared to the length of the chamber. The calculation was done for the more general case when there is a vertical displacement in the anode wire. We did not consider transverse displacement of the anode wire.

The two-dimensional grid of image charges is shown in Fig. 2, where the displacement of the anode wire is denoted by δ . The potential in the cell can be evaluated by summing the contribution from all the charges. The two-dimensional induced charge profile on a continuous cathode plane can be calculated from the electrostatic potential. After integrating the charge distribution along the y direction, the charge profile as a function of x is given by

$$\frac{dQ}{dx} = \frac{1}{2\pi a} \sum_{m=0, \pm 1, \dots} \sum_{n=0, \pm 1, \dots} \frac{(-1)^{m+n} \left(n - \frac{1}{2}\right)}{X_\delta^3} \times \left[\frac{m + \frac{1}{2}}{\sqrt{1 + \left((m + \frac{1}{2})/X_\delta\right)^2}} - \frac{m - \frac{1}{2}}{\sqrt{1 + \left((m - \frac{1}{2})/X_\delta\right)^2}} \right], \quad (1)$$

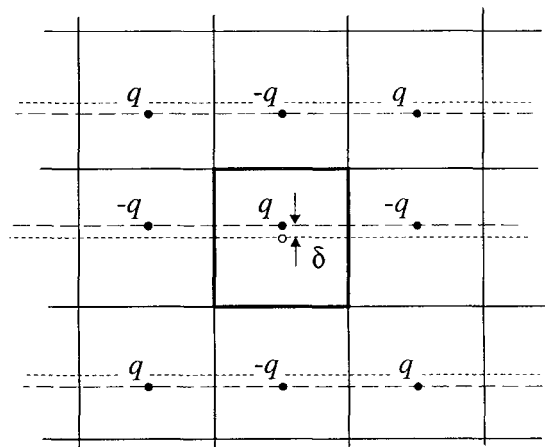


Fig. 2. Image charge grid for a vertically displaced charge q in a square cell.

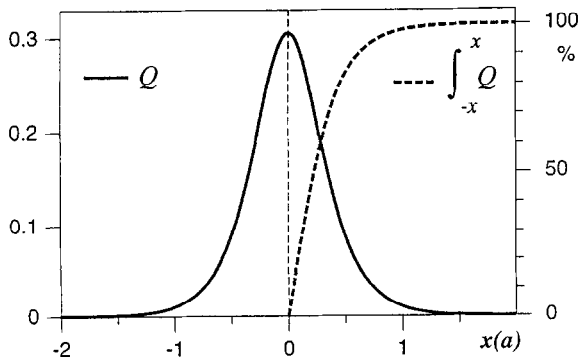


Fig. 3. Model prediction for the charge induced on the cathode (solid line) and integral from $-x$ to $+x$ (dashed line) as a function of x (along the wire).

where

$$X_n = \sqrt{\bar{x}^2 + \left(n - \frac{1}{2} + (-1)^n \bar{\delta}\right)}.$$

$\bar{x} = x/a$ and $\bar{\delta} = \delta/a$ are the x coordinate and the wire displacement in units of the cell size a , respectively. The charge is normalized to that of a unit point charge at the wire. The charge profile given by (1) has been evaluated numerically by truncating the sum at $N=200$. It has been checked that the total induced charge for $N=200$ is within 0.1% of the exact value of 0.25. The charge profile for the cell geometry identical to that of the prototype is shown in Fig. 3. As one expects, the charge profile is bell-shaped with a half-width of about $a/2$. Shown in Fig. 3 as a dashed line is the cumulative charge from $-x$ to $+x$. Over 98% of the induced charge is contained within $\pm a$. This confirms the suggestion that the optimal strip segmentation should be of order $a/2$, which would distribute the image charge among several strips.

2.3. Strip charges

From the charge distribution we have calculated the charge induced on the strips as a function of the

position of the avalanche. For this calculation the 1 mm gap between the 4 mm wide strips is ignored, and the charge on the strip is the integral of the charge profile given by (1) between the strip edges. The charge induced on individual strips depends on the location of the avalanche. Shown in Fig. 4 are two extreme cases for a strip pitch w equal to half the cell size a . Fig. 4a shows the charge profile when the avalanche is at the center of a strip. As one expects, the center strip has the most significant charge (about 54% of the total), whereas the two neighboring strips have equal but lower charge (20%). The charge outside the three most significant strips is under 6%. When the avalanche occurs between two strips, the strip charge pattern is shown in Fig. 4b. The two strips nearest to the avalanche have an equal amount of charge (41.6% each). The strips adjacent to the two in the center contain about 7.5% each of the total charge. The symmetry of the strip charge patterns for the two cases ensures that most algorithms give exact results at $x=0$ and $x=w/2$. The strip charges evolve from the pattern in Fig. 4a to that of Fig. 4b as the avalanche position moves from $x=0$ to $x=w/2$. The charge pattern for the next half-period ($x=w/2$ to $x=w$) is a mirror image of the range from $x=0$ to $x=w/2$. Then the pattern repeats with a period w . Therefore, we will only consider the range $x=0$ to $x=w/2$ in this paper.

3. Data

3.1. Detector and track data

The experimental data used for this study were obtained in a CSC prototype. Details of the experiment and results are given in Ref. [3]. A schematic of the detector is shown in Fig. 5. The prototype consists of three tracking layers, spaced 25 mm apart, each composed of one layer of Iarocci tubes and a cathode strip plane facing the open side of the profiles. The chambers are of standard design, eight 1 cm×1 cm cells in each chamber, but the extruded plastic profiles are graphite-coated with an average resistivity of 40

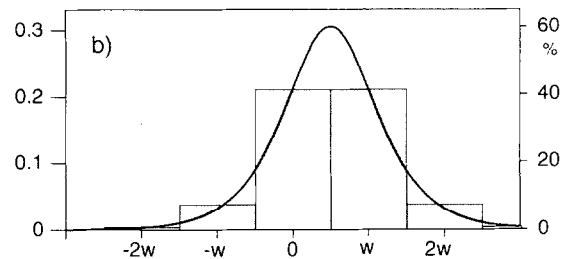
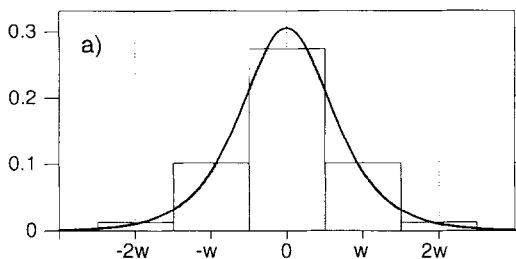


Fig. 4. Charge profile (solid line) and fraction of charge (bars, right axis) seen on strips of width $a/2$ for (a) an avalanche located in the center of a strip and (b) located between two strips.

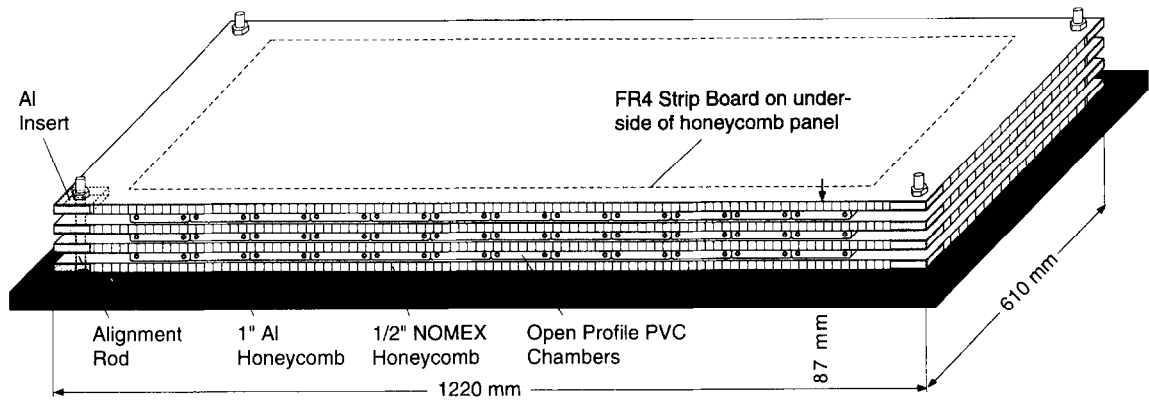


Fig. 5. Side view of the CSC prototype.

$\text{k}\Omega/\square$. This is considerably lower than that of a typical chamber for limited streamer mode applications, but is appropriate for proportional mode operation. The copper strips are photoetched onto a large ($50\text{ cm} \times 100\text{ cm}$) printed circuit board by standard techniques with an overall tolerance of $25\text{ }\mu\text{m}$. The strip pitch was found to be constant within $1\text{ }\mu\text{m}$ by surveying the boards. The average variation of the strip widths is under 1%. The chambers were operated in the proportional mode with a typical gas gain of 5×10^5 .

The prototype was tested with cosmic ray muons at the Texas Test Rig (TTR) and in a high energy muon beam (190 GeV) at CERN. The data were taken at several voltages, for two gas mixtures, and muon incident angles up to 20° . The data have been analyzed with the WCOG method which is described in Section 4.2. The resolutions are in good agreement with a simple model prediction assuming an intrinsic resolution of about $35\text{ }\mu\text{m}$. The ultimate resolution was limited by the electronics noise to about $45\text{ }\mu\text{m}$ at the highest voltage.

3.2. Comparison of data and model prediction

For comparison throughout this paper we will use a cosmic muon data set taken at 3200 V, containing about 5000 reconstructed muon tracks [3]. The incident angle of the tracks is uniformly distributed over $\pm 8^\circ$. The minimum momentum of the muons is 1.5 GeV. We choose this data set because the angular spread of the cosmic muons randomizes the correlation of the systematic errors in the three layers. The muon passes through each layer at a different relative strip position, resulting in different systematic errors. The residual reveals clearly the systematic error in each layer. The beam muons, in contrary, are almost perpendicular to the chamber and traverse each layer at the same strip position. The systematic errors are the same for all three points; the track gets shifted, but the residual is not affected by the systematic error.

The strip charges, corrected for gain and pedestal, are shown in Fig. 6 (points) as a function of the distance from the strip center x . The x -position of charge centroids was calculated using the WCOG algorithm. Shown in Fig. 6 is a x -range of half a strip width (2.5 mm); $x=0$ corresponds to the configuration of Fig. 4a (avalanche in center of strip) and $x=w/2$ to that of Fig. 4b (avalanche between two strips). The solid lines are the predictions from our model; Q_1 is the charge of the most significant strip, and Q_2 and Q_3 are the charges of neighboring strips, with the convention that $Q_1 > Q_2 > Q_3$. All charges are expressed as a fraction of the sum of Q_1 to Q_4 . The agreement between data and prediction is very good, except for a

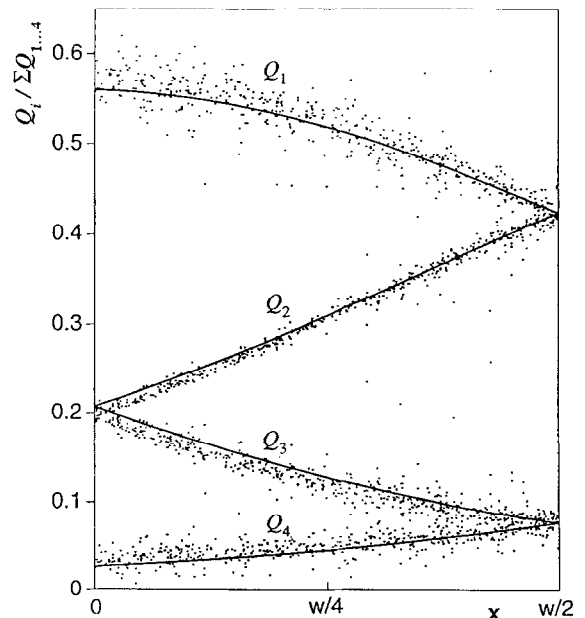


Fig. 6. Strip charges for the reference data set (points) and their model prediction (lines) as a function of centroid position.

small discrepancy around $x=0$ for Q_2 and Q_3 . This is due to ignoring the gap between the strips and a $10 \times 10 \text{ mm}^2$ cell in the model. Simulations using a more realistic cell size of $9 \times 11 \text{ mm}^2$ and taking the 1 mm gap between the strips into account improved the agreement between data and model. The systematic errors are not sensitive to such differences. Therefore, we compare the data only to the $10 \times 10 \text{ mm}^2$ cell, no-gap model.

4. Centroid-finding algorithms

We use the term “centroid” to denote the best estimate for the avalanche position by a given algorithm. There are three broad classes of algorithms commonly used to determine the centroid of CSCs: The center of gravity method, the ratio method, and the use of empirical functions.

For this investigation the centroids in the top and bottom layers were reconstructed using the WCOG algorithm (see Section 4.2), corrected for the systematic error (Section 5). The WCOG algorithm was found to have the least dependence on the incident angle and on displacements of the anode wire (see Section 7). The algorithm under investigation was applied to the middle layer. The residual r_M , in this case, was defined as the distance of the middle layer centroid from the line drawn between top and bottom corrected centroids. The error for this residual is

$$\sigma_M = \sqrt{\frac{1}{2} \sigma_{t,b}^2 + \sigma_a^2},$$

where $\sigma_{t,b}$ are the measurement errors of the top and bottom layers, respectively, and σ_a is the resolution of the middle layer when using the algorithm under study; σ_M , therefore, provides a relative measure of σ_a , and is, together with its correlation with r_M , used to compare the different algorithms.

To obtain the single layer resolution, σ_L , the same algorithm was applied to all three (identical) layers. A straight line fit, in this case, results in the correlated residuals $\pm r$, $\mp 2r$ and $\pm r$ for the three layers, respectively. The σ of the r -distribution, σ_r , and σ_L are related by:

$$\sigma_L = \sqrt{6} \cdot \sigma_r.$$

4.1. Center of gravity (COG)

The standard center of gravity (COG) algorithm calculates the centroid X_n by weighting each strip position x_i with its charge Q_i , either for a predetermined number of strips, n , or for strips above a certain threshold (bias)

$$X_n = \frac{\sum_{i=1}^n x_i Q_i}{\sum_{i=1}^n Q_i}. \quad (2)$$

The bias is usually computed as a fraction of the total charge to avoid the event-by-event Landau fluctuations, or is adjusted for a baseline shift caused by common-mode noise in all strips [12]. The bias can be determined by the counts outside the peak of the charge distribution.

Based on our model we calculated the systematic error for the simple COG algorithm using a fixed number of strips. The results are shown in Fig. 7 as a function of the true avalanche position. As one can see, the systematic error of the 3-strip COG (X_3) reaches $600 \mu\text{m}$ when the avalanche approaches the strip boundary at $x=w/2$, then the sign of the systematic error changes, and X_3 decreases to zero at the center of the next strip. This behavior is, in fact, a general feature for all COGs based on an odd number of strips, even though the maximum amplitude of the systematic error decreases as the number of participating strips increases. The discontinuity exists also for COGs based on a fixed even number of strips. For example, the 4-strip COG (X_4) is continuous at $x=w/2$, but has a discontinuity at $x=0$ and $x=w$.

This peculiarity of the COG method is illustrated in Fig. 8a–Fig. 8c, where the residual r_M is displayed for different bias levels using data. The scatter plots on the left show r_M in correlation with the centroid position for an x range of about six strip pitches; the center graph is a histogram of r_M , and the plots to the right show a comparison of data and model (solid line) over one pitch of the scatter plot on the left.

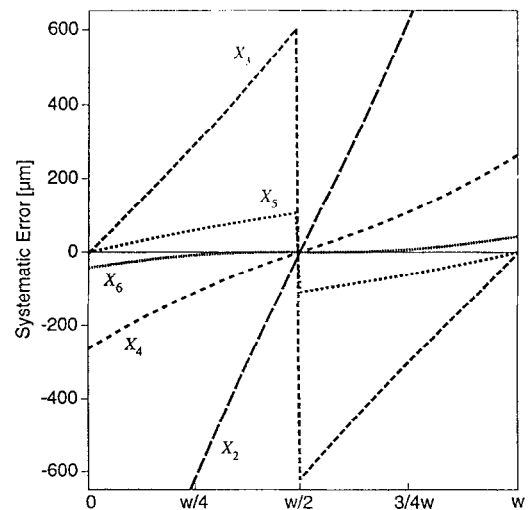


Fig. 7. Systematic difference between COG centroid and true avalanche position as a function of distance from the strip center.

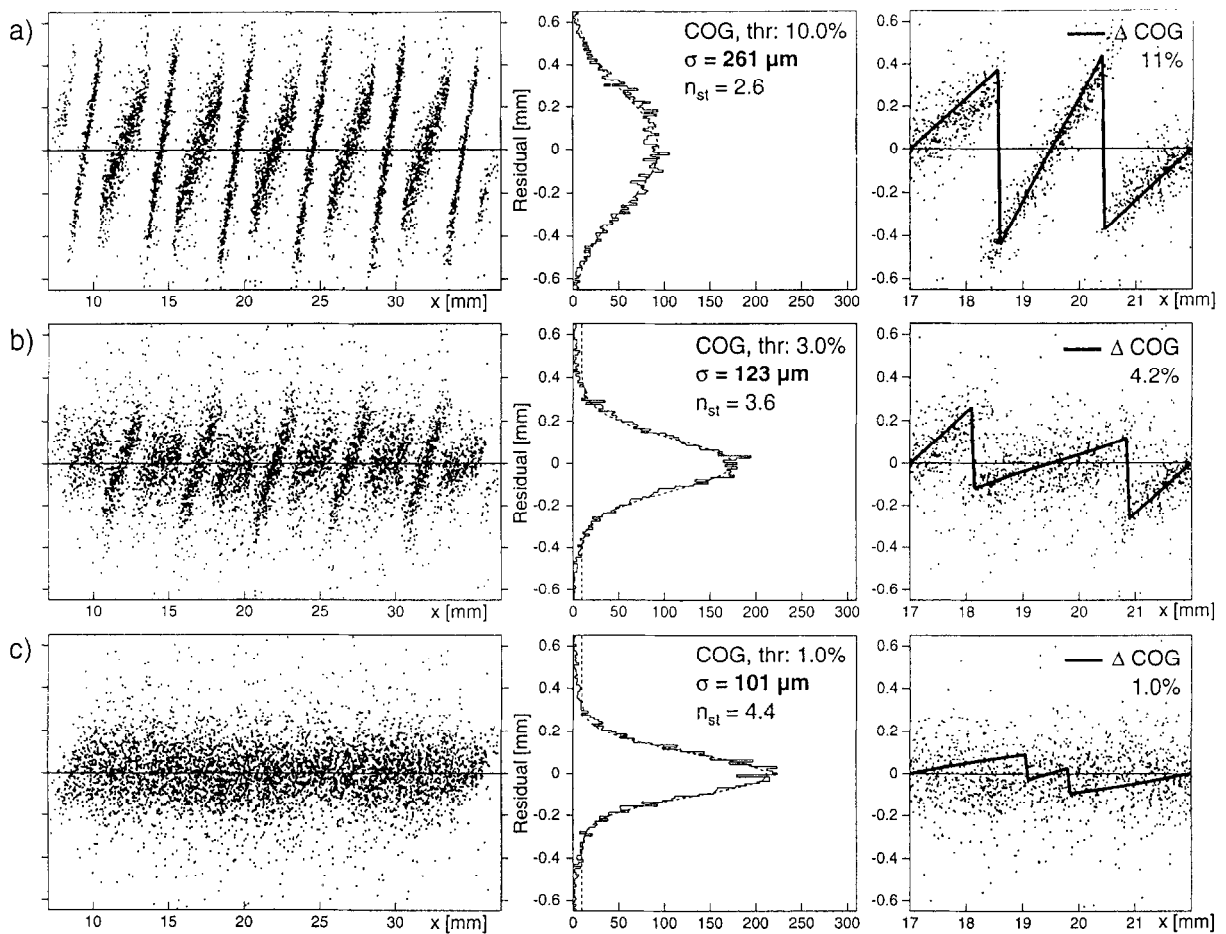


Fig. 8. Residuals as a function of middle layer centroid (left), residual distribution and result of a Gaussian fit (center), and enlarged scatter plot with error prediction (right) for various threshold settings (a)–(c) using the COG method.

The graphs for a relatively high threshold of 10% are shown in Fig. 8a. When the avalanche position is near the center of a strip ($x=17, 22$ mm, see enlarged graph on the right of Fig. 8a), the residual follows the curve labeled X_3 in Fig. 7, which is the systematic error for the 3-strip COG. As the avalanche position approaches the edge of the strip ($x=19.5$ mm), the threshold only allows two strips. The data now switch over to X_2 (see Fig. 7), creating a discontinuity. Exactly this behavior is reproduced by our model if we assume a 11% bias. The predicted systematic error is shown as solid lines in the right graphs of Fig. 8. This behavior persists as the bias is lowered even though the size of the discontinuity decreases. In Fig. 8b the threshold is 3.0%. The residuals follow X_4 when they are near the edge of the strip and X_3 when they are less than 1 mm from the center of a strip. Our model predicts a similar behavior for a 4.2% threshold. When the bias is very low, for example 1% as shown in Fig. 8c, as many as five strips are above the threshold for

most of the range. The data follow X_5 initially, then switch to X_4 near the edge, also in good agreement with our model prediction.

As the number of strips used to calculate the centroid increases, so does the noise. The average number of strips above threshold, n_{st} , is shown in the center graphs of Fig. 8. The periodic structure of the systematic deviation is clearly visible in the scatter plots if $n_{st} \leq 4$. When more than four strips are used, all structure disappears (Fig. 8c), and a correction of the systematic error, which is still on the order of $100 \mu\text{m}$, becomes impossible.

A measure for the resolution achievable is the σ of the distribution of residuals shown in the histograms in the center of Fig. 8. The poor resolution of the simple COG algorithm results from the large excursions and discontinuities in the systematic error. The discontinuity makes a systematic correction impractical. Without correction, the best σ achieved is $101 \mu\text{m}$, corresponding to the case of a 1% bias in Fig. 8c.

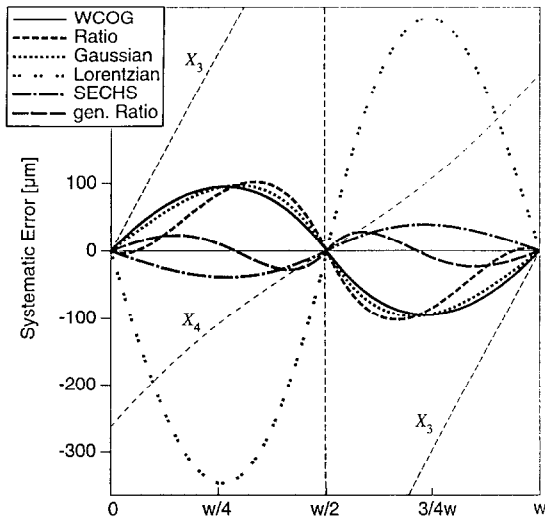


Fig. 9. Predicted systematic error for various centroid finding algorithms as a function of the centroid x -position. x is measured in strip widths w from the center of a strip.

4.2. Weighted center of gravity (WCOG)

The weighted center of gravity (WCOG) method is motivated by the observation that the systematic error of X_3 vanishes at $x=0$, and that of X_4 is zero at $x=w/2$. This is a direct consequence of the symmetry of the charge pattern shown in Fig. 4. The spatial

resolution can be improved by using a linear combination of X_3 and X_4 such that the algorithm approaches X_3 at $x=0$, and evolves continuously to X_4 at $x=w/2$. The WCOG centroid X_{wcoG} is a weighted mean of the 3-strip and 4-strip COGs,

$$X_{\text{wcoG}} = X_{\text{wcoG}} X_3 + \left(1 - \frac{2X_{\text{wcoG}}}{a}\right) X_4,$$

where the weighting factors for X_3 and X_4 are X_{wcoG} and $1 - 2/a X_{\text{wcoG}}$, respectively. X_{wcoG} can be solved in terms of X_3 and X_4 to yield

$$X_{\text{wcoG}} = \frac{w}{2} \left(\frac{X_3}{w/2 + (X_3 - X_4)} \right). \quad (3)$$

The systematic error for X_{wcoG} is shown in Fig. 9 (solid line). Indeed, the systematic deviation is continuous, vanishing at $x=0$ and $x=w/2$. The amplitude of $90 \mu\text{m}$ is much smaller than that of X_3 and X_4 . The data, analyzed with the WCOG algorithm and presented in a similar format as the COG method (without correcting for the systematic error), are shown in Fig. 10a. The residual distribution for the WCOG method has a σ of $109 \mu\text{m}$, slightly worse than the 1% bias COG method; the predicted error agrees well with the data. The systematic error of X_{wcoG} has been Fourier analyzed, and the coefficients have been used to correct the position of the centroids (see Section 5). Fig. 10b shows the *corrected* distributions for the

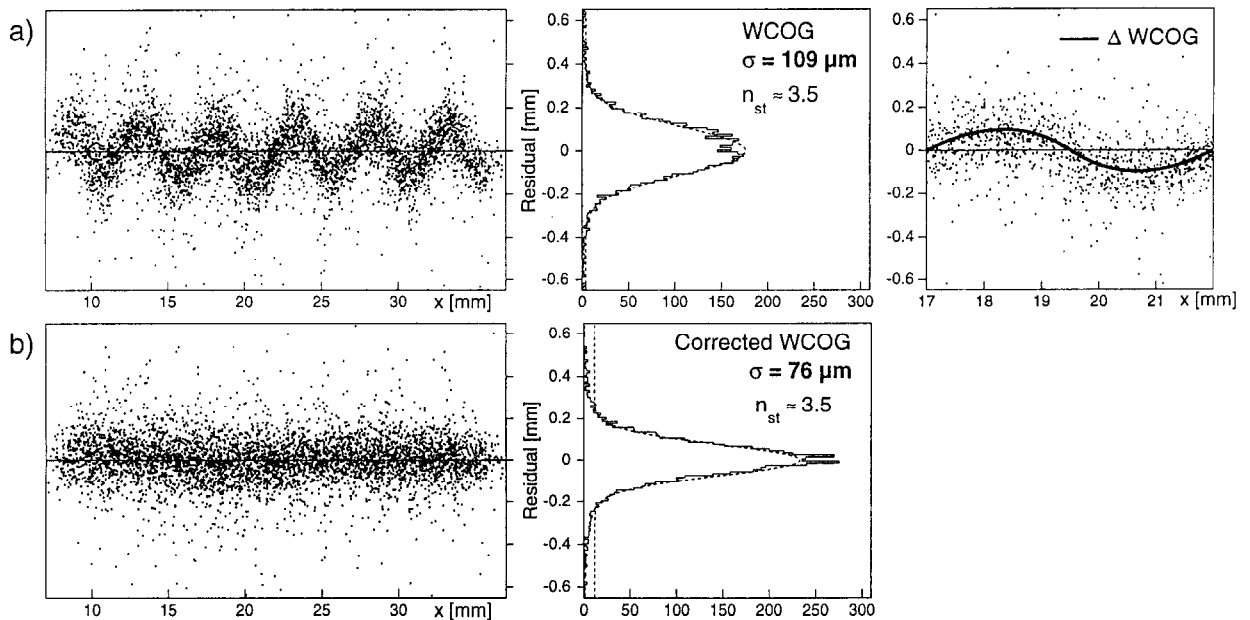


Fig. 10. Residuals as a function of the middle layer centroid (left), residual distribution and result of a Gaussian fit (center), and enlarged scatter plot with error prediction (right) for the WCOG method before (a) and after (b) correcting for the systematic error.

residuals: the scatter plot on the left exhibits no noticeable periodic structure, and the σ of the residual improves by more than 30% to 76 μm . This is significantly better than the best result obtained by the standard COG method.

4.3. Ratio method

This method [10,2] is based on the ratios of charges to the left, $f_L = Q_3/Q_1$, and to the right, $f_R = Q_2/Q_1$, of the peak charge Q_1 . It uses the information from only three strips. The centroid for the ratio method X_{ratio} is defined as

$$X_{\text{ratio}} = \frac{w}{2} \left(1 - \frac{1-f_R}{1-f_L} \right) = \frac{w}{2} \left(\frac{Q_2 - Q_3}{Q_1 - Q_3} \right). \quad (4)$$

When the avalanche occurs at the center of a strip ($x=0$), $Q_2=Q_3$, resulting in $X_{\text{ratio}}=0$. When the avalanche occurs between the strips ($x=w/2$), $Q_1=Q_2$, resulting in $X_{\text{ratio}}=w/2$. Therefore, the ratio algorithm also has continuous systematic error which is shown in Fig. 9 (short dashed line) as predicted by our model. The comparison between data and prediction using this method is shown in Fig. 11, again before a) and after b) a Fourier based correction is applied. There is also good agreement between data and theory. Even though the σ of the residual distribution has been improved from 128 to 81 μm by applying the correction, the periodic structure is still evident after correc-

tion (Fig. 11b). One explanation for this behavior (as shown in Section 7.1) is the sensitivity of the ratio method's systematic error on the incident angle. The non-normal tracks require a different set of Fourier coefficients to be fully corrected. The correction used in Fig. 11b is based on normal incidence.

4.4. Generalized ratio method

The ratio method can be generalized to a one-parameter function of f_L and f_R . Following the notation of [1], an angle α can be defined as

$$\alpha = \tan^{-1} \left(\frac{1-f_L}{1-f_R} \right).$$

α increases from $\pi/4$ to $\pi/2$ as the avalanche moves from the center to the edge of a strip. The following function, first used in Ref. [1], determines the centroid

$$X_{\text{gRatio}} = \frac{w}{2} \left(\frac{\tan^{-1}(b(\alpha - \pi/4))}{\tan^{-1}(b\pi/4)} \right), \quad (5)$$

which is exact at $x=0$ and $x=w/2$ for an arbitrary constant b . The value of b can be varied to minimize the systematic error in the range $0 < x < w/2$. Using our model, $b=1.65$ was found to give the smallest overall amplitude, as shown in Fig. 9 (long-dashed line). When applied to our data, the generalized ratio method gives slightly better corrected results (79 μm) than the ratio-method introduced in Section 4.3, but otherwise is

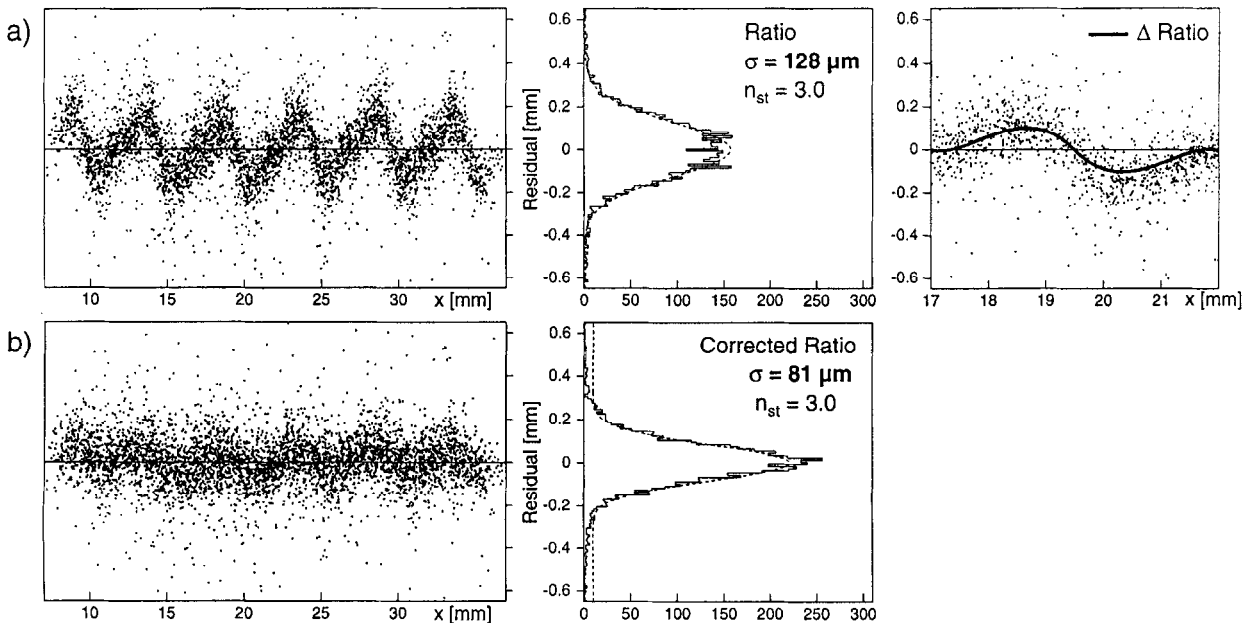


Fig. 11. Residuals as a function of the middle layer centroid (left), residual distribution and result of a Gaussian fit (center), and enlarged scatter plot with error prediction (right) for the ratio method before (a) and after (b) correcting for the systematic error.

very similar. It also exhibits a non-vanishing periodicity in the corrected scatter plot (not shown).

4.5. Empirical functions

The basic idea in this approach is to match the observed charge distribution to an analytic function with a small number of parameters. The minimum number of parameters is three; a_1 for the overall normalization, a_2 for the position of the centroid, and a_3 for the width of the charge distribution. The spatial resolution is expected to be degraded by the use of more than three strips in the fitting procedure. We therefore confine our analysis to 3-parameter functions, that can be solved analytically and give exact results at $x=0$ and $x=w/2$. The following are some examples.

Gaussian

The charge profile is approximated by the familiar function

$$Q_i = a_1 e^{-(x_i - a_2)^2 / a_3^2}.$$

The results of a fit of a Gaussian to the charge profile based on our model is shown in Fig. 12. The agreement is acceptable. The centroid a_2 can be determined as

$$a_2 = \frac{w}{2} \left(\frac{\ln Q_1 / Q_3 - \ln Q_1 / Q_2}{\ln Q_1 / Q_3 + \ln Q_1 / Q_2} \right). \quad (6)$$

The systematic error for the Gaussian algorithm is shown in Fig. 9 (dotted line). The error for this method is similar to that of the WCOG algorithm. The comparison with the data is not repeated here.

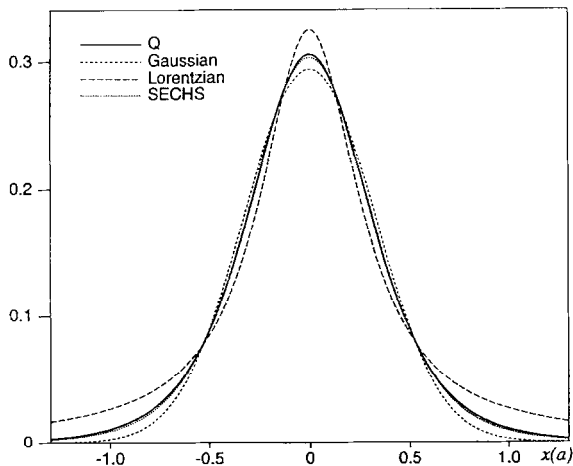


Fig. 12. Charge profile for a square cell of size a (solid line), as predicted by our model, and results of fits with 3-parameter empirical functions.

Lorentzian

This function, also known as the Cauchy or the Breit–Wigner formula, describes the charge profile as follows:

$$Q_i = \frac{a_1^2}{(x_i - a_2)^2 + a_3^2}.$$

The Lorentzian function is the poorest approximation of the charge profile (see Fig. 12). The centroid is

$$a_2 = \frac{w}{2} \left(\frac{Q_1 Q_2 - Q_1 Q_3}{Q_1 Q_2 + Q_1 Q_3 - 2Q_2 Q_3} \right). \quad (7)$$

The systematic error is also shown in Fig. 9 (double dotted line). This algorithm exhibits the largest error and is therefore not pursued further.

Hyperbolic secant squared (SECHS)

This relatively simple 3-parameter function [13–15],

$$Q_i = \frac{a_1}{\cosh^2(\pi(x_i - a_2)/a_3)},$$

is a very good approximation to the charge profile used in our model (Fig. 12, dotted line). The difference between the model and the function is hardly noticeable. The slightly more complicated solution for the SECHS centroid is

$$a_2 = \frac{a_3}{\pi} \tanh^{-1} \frac{\sqrt{Q_1/Q_3} - \sqrt{Q_1/Q_2}}{2 \sinh(\pi w/a_3)}. \quad (8)$$

with

$$a_3 = \frac{\pi w}{\cosh^{-1} \frac{1}{2} (\sqrt{Q_1/Q_3} + \sqrt{Q_1/Q_2})}.$$

The systematic error for the SECHS algorithm is shown in Fig. 9 (dash dotted line). This algorithm has one of the smallest error amplitudes ($40 \mu\text{m}$) and can be corrected adequately by a single term Fourier series. When applied to the data, the uncorrected residual distribution has already a σ of $84 \mu\text{m}$ (Fig. 13a). After correction, no periodic structure remains in the scatter plot and the σ becomes $74 \mu\text{m}$, the smallest of all the algorithms studied. This is the best centroid-finding algorithm we have studied so far.

5. Systematic correction

We have computed the systematic deviations for all centroid-finding algorithms described in Section 4. They are shown in Fig. 9. The systematic error can be parametrized by a Fourier series to provide a point-to-point correction for the algorithmic centroid. The

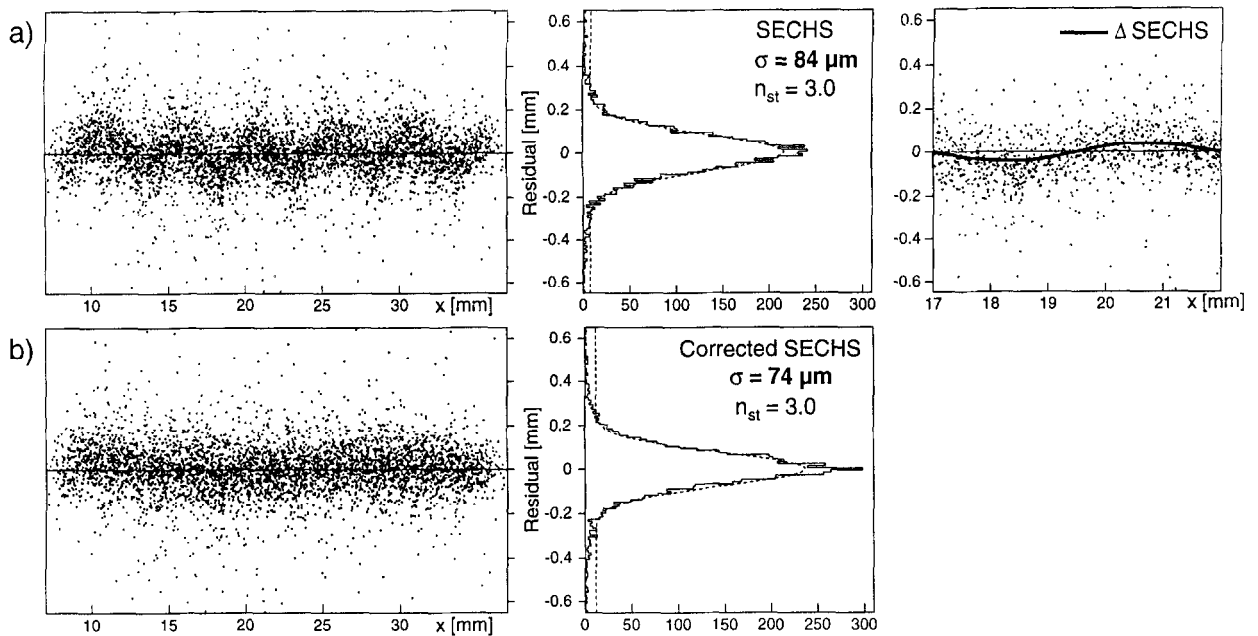


Fig. 13. Residuals as a function of the middle layer centroid (left), residual distribution and result of a Gaussian fit (center), and enlarged scatter plot with error prediction (right) for the SECHS method before (a) and after (b) correcting for the systematic error.

systematic deviation for all algorithms was analyzed using the following function

$$\Delta(x) = \sum_m a_m \sin\left(\frac{2m\pi}{w}x\right).$$

The coefficients a_m for the first six terms are listed in Table 1 for each algorithm. As one can see, the convergence is quite rapid for the WCOG and SECHS methods; all terms are less than 1% for $m > 3$, whereas both ratio-methods require more coefficients to reach the same accuracy. Among the algorithms, the SECHS method has the smallest amplitude and converges most rapidly.

The centroid, X_u , calculated by a certain algorithm, was corrected in practice by a two step procedure: First, the Fourier series $\Delta_u(x)$ for this algorithm, usually ignoring all terms $< 1\%$, was used to estimate x_{est} :

$$X_{est} = X_u - \Delta_u(X_u).$$

Then the estimated x is used to re-calculate the corrected centroid

$$X_u^{corr} = X_u - \Delta_u(x_{est}).$$

6. Summary

Table 2 lists, for all studied algorithms, the σ obtained by a four parameter Gauss-fit to the distribution of residuals. Columns 2 and 3 list σ_M as reported in the center graph of Fig. 8, Fig. 10, Fig. 11 and Fig. 13, uncorrected in column 2 and corrected in column 3. Columns 4 and 5 list the single layer resolution σ_L , obtained by applying the same algorithm to all three layers and correcting for the systematic error. Column 4 is for the reference data

Table 1

Fourier coefficients a_m (in μm) of the predicted systematic errors for various centroid finding algorithms

m	WCOG	Ratio	gRatio	Gauss	Lorentz	SECHS
1	97.73	85.17	4.19	97.34	-351.82	-39.43
2	-4.47	-37.57	23.87	-13.98	12.68	2.52
3	3.51	0.49	-7.18	4.25	-5.12	-0.72
4	-0.57	-5.16	2.84	-1.81	2.15	0.30
5	0.76	-0.02	-1.41	0.93	1.11	-0.15
6	-0.17	-1.56	0.80	-0.54	0.65	0.09

Table 2

Uncorrected and corrected residual resolutions for various centroid finding algorithms, fitting methods and data sets

Method	Resolution [μm]			
	σ_M	σ_M^{COFF}	σ_L^{COFF}	σ_L^{COFF}
	Cosmics 3.2 kV		Beam 3.3 kV	
COG, thr: 6%	322	–	–	–
COG, thr: 3%	123	–	–	–
COG, thr: 1%	101	–	–	–
WCOG	109	76	62.3	52.0
Ratio	128	81	63.9	49.5
Gen. Ratio	82	79	64.1	49.5
Lorentz	–	–	–	53.8
SECHS	84	74	59.3	45.8

(cosmics at 3200 V and $\pm 8^\circ$ incidence angle) and column 5 is for beam muons at 3300 V. Due to the better signal to noise ratio at the higher voltage and the almost normal incidence ($\pm 2.5^\circ$), the beam data have better resolutions.

7. Sensitivity to other effects

7.1. Angle of incidence

The spatial resolution in a CSC is a strong function of the incident angle of the charged track. This is due to the additional error arising from the fluctuation in the ionization process. The angle dependent error can be approximated by

$$\sigma_\theta = \frac{a \tan \theta}{\sqrt{12N}},$$

where N is the number of primary clusters, corrected for path length for inclined tracks. It is therefore of interest to compare the angle dependence of the systematic deviation for various algorithms.

This behavior has been simulated by uniformly distributing a fraction of the original charge over a distance $a \tan \theta$ along the wire. The resulting charge profile widens slightly. The changes in systematic error caused by tracks of 5° , 10° and 15° incident angle are shown in Fig. 14, Fig. 15, Fig. 16, and Fig. 17, left side, for the different algorithms. Both ratio methods show a large (30–100%) dependence on the angle; the WCOG method shows none and the SECHS method shows only about a 10% dependence.

This model prediction is borne out by the data. For the almost normal tracks in the beam data the ratio methods have a better single layer resolution σ_L than the WCOG, whereas for the $\pm 8^\circ$ sample in the cosmic data, the ratio methods are worse than the WCOG.

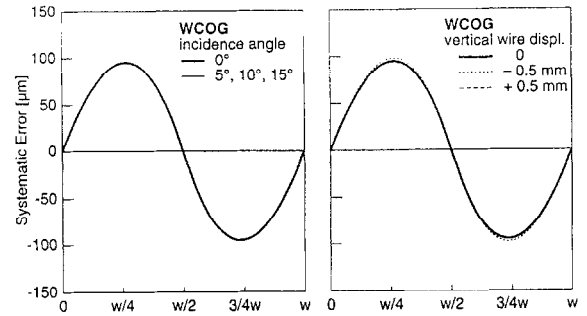


Fig. 14. Systematic error for the WCOG method at various incident angles (left) and for vertical displacement of the anode wire (right), as predicted by our model.

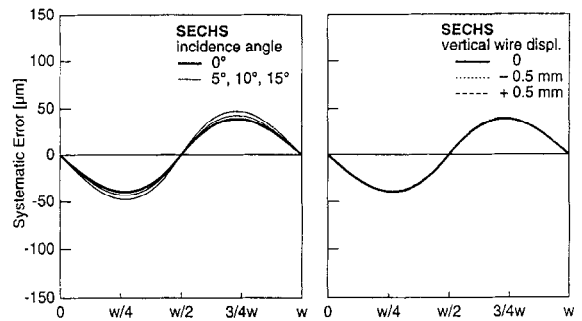


Fig. 15. Systematic error for the SECHS method at various incident angles (left) and for vertical displacements of the anode wire (right), as predicted by our model.

The SECHS algorithm, having a small error to start with and showing only a small angular dependence, delivers the best resolution in all cases, 7–9% better than all other algorithms.

7.2. Mechanical tolerance

One of the advantages of the CSC technique is its low dependence on the placement accuracy of the

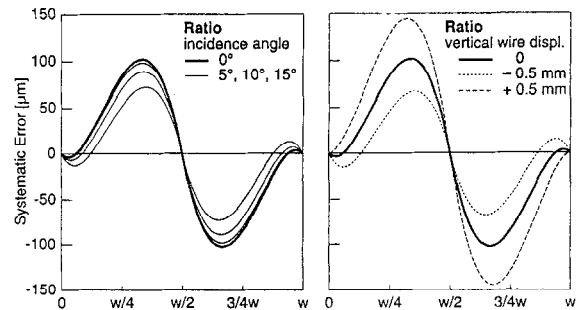


Fig. 16. Systematic error for the Ratio method at various incident angles (left) and for vertical displacements of the anode wire (right), as predicted by our model.

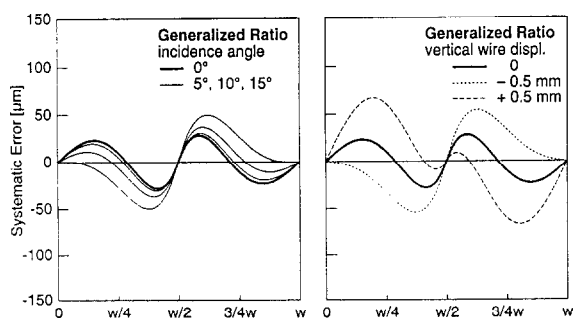


Fig. 17. Systematic error for the Generalized Ratio method at various incident angles (left) and for vertical displacements of the anode wire (right), as predicted by our model.

anode wire. The charge profile is insensitive to a lateral misplacement of the anode wire, but is more sensitive to up–down displacements. Shown in Fig. 14, Fig. 15, Fig. 16, and Fig. 17, right side, are the systematic errors resulting from a vertical displacement of $+0.5$ and -0.5 mm, simulated in our model. As one can see, the effects are very small in WCOG and SECHS methods, but are very large for the Ratio methods.

8. Conclusion

Considerable progress has been made in the understanding of cathode strip chambers in the last few years. It becomes increasingly clear that the centroid-finding algorithm plays a significant role in achieving the best possible resolution for a given CSC geometry. We showed, with the best available data, that at the level of $50\text{ }\mu\text{m}$, only algorithms using a small number of strips with continuous and small systematic error give good results. Several algorithms were found to satisfy this condition, and gave comparable results. The standard COG method does not meet the criteria, but a slight modification, the WCOG method, gives

good results, especially under real experimental conditions. Another algorithm (SECHS) was found to improve our previously published best single layer resolution for beam muons by $7.5\text{ }\mu\text{m}$ to $45.8\pm 0.7\text{ }\mu\text{m}$. It is remarkable that a simple static point charge model gives a very good description of the data. This simple model can assist us in the design and further optimization of CSCs in the future.

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